

NEURO 610: Cognitive Neuroscience and Functional Neuroimaging

Term: Fall 2026 | Instructor: Dr. Brenda Milner (bmilner@med.neuro.edu)

Course Description & Overview:

Principles of functional magnetic resonance imaging (fMRI), statistical parametric mapping (SPM), and neural decoding architectures.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
fMRI Preprocessing & GLM Analysis Lab	25%	Week 03	SPM Pipeline Script & Writeup
Resting-State Functional Connectivity Project	30%	Week 07	Jupyter Notebook & Paper
Neuroimaging Research Grant Proposal	35%	Week 11	12-Page NIH Format Proposal
Journal Club Presentation	10%	Week 05	Slide Deck & Oral Defense

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: BOLD Contrast Biophysics	Huettel (Ch. 1–2); Ogawa et al. (1990)
Week 02	Module 02: Experimental Paradigm Design	Huettel (Ch. 4 & 5)
Week 03	Module 03: Spatial Normalization & SPM12 Architecture	SPM Documentation
Week 04	Module 04: General Linear Model & Inference	Friston et al. (1995)
Week 05	Module 05: Multi-Voxel Pattern Analysis (MVPA)	Haxby et al. (2001); Norman et al. (2006)
Week 06	Module 06: Diffusion Tensor Imaging & Tractography	Mori & Zhang (Ch. 3)
Week 07	Module 07: Resting-State Network Architectures	Raichle (2001); Power et al. (2011)
Week 08	Module 08: Magnetoencephalography & Source Analysis	Hari & Puce (Ch. 6–8)
Week 09	Module 09: Optical Neuroimaging (fNIRS) Protocols	Scholkmann et al. (2014)
Week 10	Module 10: Deep Brain Stimulation & Neuromodulation	Lozano et al. (2019)
Week 11	Module 11: Machine Learning Decoders for Brain States	Haynes & Rees (2006)
Week 12	Module 12: Grant Proposal Presentations	Neuroimaging Review Guidelines